8 Beats and dispersion

8.1 Beats

Piano tuners must tune strings to an exquisitely defined frequency. To do this, they exploit an acoustic phenomenon called **beating**. When two frequencies that are very closely matched, but not quite identical, are played together, the effect is that of a single tone, but with a fluctuating amplitude. The fluctuation gets slower as the frequencies are brought closer together, until it disappears altogether when the frequencies are the same. This is literally a case of fine tuning.

To describe this mathematically, we need to add two signals that are identical except for a slight difference in frequency[1] We will also allow for a difference in wavelength, which we will discuss later. For now let us say that if the propagation speed is independent of frequency, then $v = \omega/k$ implies that if the frequencies are different, so are the wavelengths. So the signals that we are adding are:

[1] To simplify matters, we will assume that the two signals have the same amplitude, are initially in phase, and travel the same distance, so that they remain in phase when they arrive at the detector.

$$\tilde{y}_1 = Ae^{i(\omega_1 t - k_1 x)} \tag{8.1a}$$

$$\tilde{y}_2 = Ae^{i(\omega_2 t - k_2 x)} \tag{8.1b}$$

We will choose ω_2 to be the higher of the two.

Let us introduce two frequencies constructed from ω_1 and ω_2: The average frequency

$$\omega_{av} = \frac{\omega_1 + \omega_2}{2} \tag{8.2}$$

and the frequency difference

$$\delta\omega = \omega_2 - \omega_1 \tag{8.3}$$

Similarly, we introduce k_{av} and δk. Writing the two signals in terms of these new quantities and simplifying, we arrive at:

$$\tilde{y}_1 + \tilde{y}_2 = 2A\cos\left(\frac{\delta\omega}{2} - \frac{\delta k}{2}\right) e^{i(\omega_{av} t - k_{av} x)} \tag{8.4}$$

This is a wave

$$e^{i(\omega_{av} t - k_{av} x)} \tag{8.5}$$

with an amplitude that is itself a wave:

$$2A\cos\left(\frac{\delta\omega}{2} - \frac{\delta k}{2}\right) \tag{8.6}$$

Since $\delta\omega \ll \omega_{av}$, the effect is a high frequency wave with a much more slowly oscillating envelope.

Figure 8.1 shows the three waves calculated at the same position, which we can call $x = 0$.

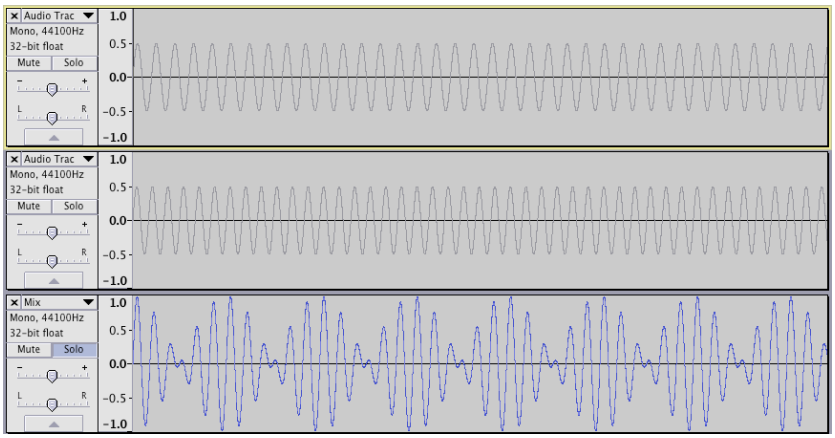


Figure 8.1: Two signals of nearly identical frequencies, and their superposition.

We hear a "wa-wa" effect, but since the loudness of a signal is a function of its intensity, that is of the amplitude squared, two half cycles of the modulated signal are heard as two separate wa's. So the angular frequency of the beats is not $\delta\omega/2$ but twice this, $\delta\omega$. In terms of frequencies, the **beat frequency** is

$$f_{beat} = f_2 - f_1 \tag{8.7}$$

This is why the beats get slower and slower as the piano tuner brings the frequencies closer together.

Beating is an interesting effect beyond piano tuning. For instance, the Doppler effect will often produce a slight frequency shift in the emitted frequency. Then the difference between the two frequencies can be measured quite effectively by mixing the signals and measuring the beat frequency, and the relative velocity of emitter and observer can be retrieved from that information. There is good evidence that bats can detect the relative velocity of prey in this way. And the same principle can be applied to measure the flow velocity in a blood vessel using the Doppler effect non-invasively.

Example 8.1: *If the bat of example 6.1 mixes the reflected signal from the insect with its own emitted signal, what is the beat frequency?*

The beat frequency is the difference between the two frequencies,

$$f'' - f = f\frac{2v_{bat}}{v - v_{bat}}$$

8.2 *Dispersion*

In the example of the previous section, the envelope described by equation (8.6) is itself a travelling wave! It travels with velocity $\delta\omega/\delta k$, which is not the same as the velocity of the component waves y_1 and y_2, which is ω/k.

So far we have only considered waves propagating in a medium where ω/k is a constant independent of k. In such a medium, waves of all wavelengths travel at the same speed, and therefore the velocity of our wave combination would the same, as $\delta\omega/\delta k = \omega/k$ in that case.

But it is not always the case that ω/k is a constant. The function $\omega(k)$ is a property of the medium. In those cases where ω/k is no longer a constant, the medium is said to be **dispersive**. Then we can no longer speak of a single propagation speed: Waves of different wavelengths travel at different speeds.

In a dispersive medium, $\tilde{y}_1$ and $\tilde{y}_2$ would travel at different speeds. The group $\tilde{y}_1 + \tilde{y}_2$ would travel at a speed that is to a good approximation $d\omega/dk$ if $\delta\omega$ and δk are indeed very small. We distinguish between the two speeds by calling the speed of a wave of a single wavelength the **phase velocity**, and the speed of a group of waves the **group velocity**[2]

[2] The terminology is confusing if we think that there ought to be a difference between *speed* and *velocity*. Unfortunately the two terms are used interchangeably in the context of wave propagation.

$$v_p = \frac{\omega}{k} \tag{8.8a}$$

$$v_g = \frac{d\omega}{dk} \tag{8.8b}$$

Let's examine the difference between v_g and v_p a little more closely. It is helpful to look at the function $\omega(k)$ for a given medium, which is known as the **dispersion relation** for that medium.

If $\omega = ck$, where c is a constant independent of the wavelength, then it's easy to see from their definitions that v_p is the same as v_g, and there is no dispersion. An example of this would be transverse waves travelling in an elastic string. We saw before that $v_p = \sqrt{T/\mu}$, which is independent of wavelength, which is the condition for no dispersion.

But consider another scenario: Surface waves in a deep body of water. It can be shown that the dispersion relation in this case is

$$\omega = \sqrt{gk} \tag{8.9}$$

The phase velocity is

$$v_p = \sqrt{\frac{g}{k}} \tag{8.10}$$

which is not a constant independent of k, so this is a dispersive medium. The group velocity is

$$v_g = \frac{1}{2}\sqrt{\frac{g}{k}} = \frac{v_p}{2} \tag{8.11}$$

In this scenario, $v_g < v_p$; this is called **normal dispersion**. Figure 8.2 shows a plot of $\omega(k)$ for deep water waves. The red line is the tangent at some value of k, so its slope is v_g for that value of k. The blue line, on the other hand, has a slope equal to v_p. Clearly the slope of the red line is smaller than that of the blue line, and this is true for any value of k we choose. So a dispersion relation $\omega(k)$ that curves downwards as in this case corresponds to normal dispersion.

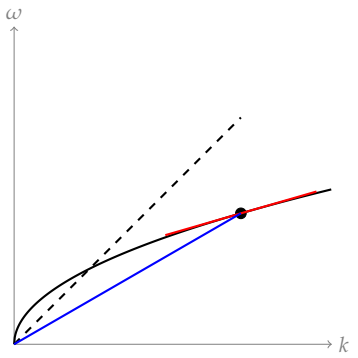


Figure 8.2: Dispersion relation for deep water waves. The dashed line represents a non-dispersive medium. The solid black curve represents $\omega(k)$; the red tangent line has a slope equal to $v_g = d\omega/dk$, and the blue line has a slope equal to $v_p = \omega/k$ for the same value of k. For this dispersion relation, $v_g < v_p$ for all k.

Is it possible to have $v_g > v_p$? Yes, and here is an instance of this. We have seen that an elastic string is a non-dispersive medium for transverse waves. But there is a hidden assumption there that is not realistic in all cases: The string is "floppy", i.e. no force is needed to bend it. In that case, the only relevant force is the tension. But if the string has some measure of stiffness, like a piano string, then the force needed to bend the string to create a transverse wave cannot be neglected. In that case, it can be shown that the dispersion relation is

$$\omega = \sqrt{\frac{T}{\mu}k^2\left(1 + \alpha^2 k^2\right)} \tag{8.12}$$

where α is a constant with dimensions of length. So for very long wavelengths with $\lambda \gg \alpha$, the correction due to stiffness is very small, and the string is non-dispersive. But for very short wavelengths $\lambda \ll \alpha$, the stiffness correction prevails, and the string is a dispersive medium.

Figure 8.3 shows a plot of $\omega(k)$ for a stiff string. Again the slope of the red tangent line is equal to v_g, while the slope of the blue line is equal to v_p. But in this case, $v_g > v_p$! This is called **anomalous dispersion**.

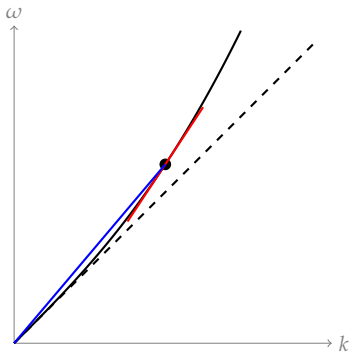


Figure 8.3: Dispersion relation for a stiff spring. The curves have the same meaning as in Figure 8.2. In this case, however, $v_g > v_p$ for all k.

Dispersion is also observed in the case of electromagnetic waves propagating in certain material media. In the case of e.m. waves, we are more familiar with the refractive index of a medium, which is defined as

$$n = \frac{c}{v_p} \tag{8.13}$$

where c is the speed of light in vacuum, and v_p is the phase velocity of a wave in a material medium.

In vacuum, $n = 1$, because $v_p = c$. Because c is a constant independent of wavelength, a vacuum is not a dispersive medium. But there are materials where v_p, and therefore n, depend on wavelength. Figure 8.4 shows plots of $n(\lambda)$ for several types of glass.

Notice that in all cases n *decreases* monotonically with λ, and therefore *increases* monotonically with k. To see what type of dispersion this corresponds to, let's express v_g in terms of n first. Because $v_p = \omega/k$, we can write the dispersion relation as

$$\omega = kv_p = \frac{ck}{n} \tag{8.14}$$

and therefore

$$v_g = \frac{d\omega}{dk} = \frac{c}{n}\left(1 - \frac{k}{n}\frac{dn}{dk}\right) \tag{8.15}$$

Since $dn/dk < 0$, it follows that $v_g < v_p$, which means that this is a case of normal dispersion.

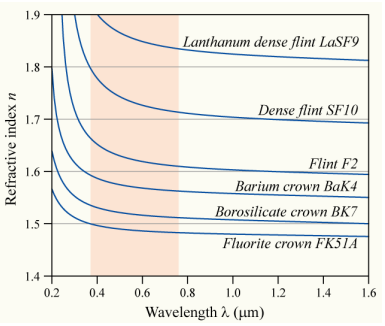


Figure 8.4: Dispersion curves for different types of glass. (Bob Marshall).